Computer Graphics I

4003-570 / 4005-761

Assignment #1: Line Drawing

Date posted: March 14th 2013

Due date: March 21st 2013

Submit to the appropriate dropbox on myCourses.

Purpose:

In this assignment, you will implement the Midpoint Line Drawing Algorithm presented in class.

Goals:

- Introduce you to implementation of 2D algorithms
- Improve your understanding of the midpoint line algorithm
- Ensure that you can install and run an application using the provided 2D programming environment on your system.

2D Programming Environment:

The programming environment that you will use for this, and the remainder of the 2D assignments, is a set of simple object-oriented classes with implementations in both Java and C++. You are free to use either of the implementations. The classes include:

- `simpleCanvas` – a simple 2D canvas that allows the ability to set a pixel.
- `Rasterizer` – a simple class that implements rasterization algorithms
- `lineTest` – the main function for the application.

You can find the files in the content area of myCourses. Look under the resources category. Note that the distribution contains folders for both C++ and Java versions of the programming environment.

The only class that you will need to modify is the `Rasterizer` class. For this assignment, you will need to complete the method `drawLine()`, implementing the Midpoint Line Drawing Algorithm. In your implementation, you need only make use of the method `setPixel()` on the passed in `simpleCanvas` object to manipulate the pixels on the canvas. The signature for the `drawLine` method in Java is:

```
void drawLine (int x0, int y0, int x1, int y1, simpleCanvas C)
```

and in C++:

```
void drawLine (int x0, int y0, int x1, int y1, simpleCanvas &C)
```

Where `(x0, y0)` and `(x1, y1)` are the endpoints of the line to be drawn. You are free to add additional members and methods to the `Rasterizer` class as you see fit, however, you cannot modify the `simpleCanvas` or `lineTest` classes.

Code from the Web:

It is common knowledge that the code for the Midpoint line algorithm is freely available on the Internet. More specifically, the applet mentioned in class contains a complete implementation. You are free to use this applet (and other web references) as a guide, but please do not simply cut and paste from this (or any other) website.

Additional Libraries:

The Java implementation of the programming environment makes use of the standard awt and Java2D classes to access the 2D drawing areas, so no additional libraries or Java files, other than the ones provided by the assignment are required.

However, since C++ does not have a standard 2D graphics environment, you will need some additional libraries if you choose to use the C++ implementation for your submission. The C++ implementation is built on the SFML (Simple and Fast Multimedia Library), a free, cross-platform and lightweight library for performing common multimedia tasks. It can be downloaded at http://www.sfml-dev.org/download.php. Version 1.6 of the libraries are used. There are precompiled binaries for Windows, Mac, and Linux. (NOTE: For Windows, there are precompiled binaries for Visual Studio 2005, and Visual Studio 2008. If you are using Visual Studio 2010, a precompiled version of the libraries is available in the content area of myCourses. Look under the resources category.

Please be sure to refer to the installation instructions for your given platform:

- Windows (Visual Studio) - http://www.sfml-dev.org/tutorials/1.6/start-vc.php
- Mac OS X -- http://www.sfml-dev.org/tutorials/1.6/start-osx.php
- Linux -- http://www.sfml-dev.org/tutorials/1.6/start-linux.php

If you are doing the assignment in C++ (and thus using the SFML libraries), you will need to link the following libraries (or frameworks for Mac/Xcode) into your application or project:

- sfml-graphics (or sfml-graphics-d for debug version in Windows)
- sfml-window (or sfml-window-d for debug version in Windows)
- sfml-system (or sfml-system-d for debug version in Windows)

Also, if you are doing the assignment in C++ for Windows using Visual Studio, be sure that the SFML dll's are placed in a folder that is in your PATH.

Submission

Your submission should only include the files for the `Rasterizer` class that you have modified. For the Java version, this would mean the file `Rasterizer.java` and for the C++ version, this means the files `Rasterizer.cpp` and `Rasterizer.h`. PLEASE DO NOT SUBMIT ANY OTHER FILES. Specifically, since you should not be changing the other classes, the source for these classes should not be submitted.

Submissions should be made to the dropbox labeled Assignment 1 - Line Drawing

Grading

This assignment is worth 10 points distributed as follows:

- 5.2 points for getting the environment to run this creating an application that runs.
- 0.6 points for each line in the test program. There are 8 such lines.

Note that the correct output for the test program should look like this: